

Emanuele Rossi

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Researcher working on machine learning for animal communication, part of a broader interest in leveraging ML to recover structure in complex systems: previously social graphs (Temporal Graph Networks, SIGN) at Twitter and Imperial College London, and generative models for biomolecules at VantAI.

Google Scholar metrics: 4,444 citations; h-index 15 (August 2026).

Education

Imperial College London Oct 2020 – Feb 2024

Ph.D. in Computer Science – Supervisor: Prof. Michael Bronstein

- Research focused on **graph machine learning methods** for large real-world systems

University of Cambridge Oct 2018 – Jun 2019

Master in Advanced Computer Science, Distinction (4.0 GPA)

Imperial College London Oct 2015 – Jun 2018

Bachelor in Computer Science, First Class Honors (4.0 GPA)

Experience

Postdoctoral Researcher Rome (remote)

GLADIA Research Group [🔗](#), Sapienza University of Rome – With Prof. E. Rodolà Oct 2025 – Present

- Developing machine learning methods for animal communication.

Machine Learning Researcher NYC (remote)

VantAI Jan 2024 – Aug 2025

- **Co-led development of Neo-1** [🔗](#), a state-of-the-art all-atom latent diffusion model for multi-modal structure prediction and de-novo generation of biomolecular complexes, trained on 100+ GPUs.

Machine Learning Researcher London

Twitter Jun 2019 – Feb 2023

- Developed **graph ML models** for Twitter’s social graph and deployed a recommendation model that generated **1.2M additional daily follows**

Machine Learning Researcher London

Fabula AI Mar 2019 – Jun 2019

- Developed graph ML models for fake news detection as part of Fabula AI’s small research team before [Twitter acquired the company](#) [🔗](#)

Software Engineering Intern California

Google Jul 2017 – Oct 2017

Selected Publications

[Full list](#)

Animal Communication & Audio

- D. Marincione, D. Crisostomi, R. Dessì, E. Rodolà, and **E. Rossi**. [Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models](#) [🔗](#). *NeurIPS Workshop on AI for Animal Communication 2025, Spotlight (top 15%)*.
- F. Mustun, C. Semenzin, R. Dessì, P. R. Guerrero, P. Orhan, A. Emanuelli, **E. Rossi et al.** Open-Whistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. *Under Review 2026*.
- **E. Rossi** and E. Rodolà. [Communicating Sound Through Natural Language](#) [🔗](#). *arXiv 2026*.

Graph Learning

- **E. Rossi et al.** [Temporal Graph Networks for Deep Learning on Dynamic Graphs](#) [🔗](#). *ICML GRL*

Workshop 2020. **1,700+ citations** and **1,000+ GitHub stars** [↗](#).

- **E. Rossi***, F. Frasca* et al. **SIGN: Scalable Inception Graph Neural Networks** [↗](#). *ICML GRL Workshop 2020*. **600+ citations** and used in **production at Airbnb** [↗](#).
- B. Chamberlain, S. Shirobokov, **E. Rossi** et al. **Graph Neural Networks for Link Prediction with Subgraph Sketching** [↗](#). *ICLR 2023, Oral (top 5%)*. **200+ citations**.

Leadership

Co-Founder, LeadTheFuture ([Forbes article](#) [↗](#))

Jun 2018 – Mar 2023

- Built a non-profit mentoring community with more than **200 mentors** and **500 selected mentees**

Supervision and Mentoring

- **Daide Marincione** – PhD Student @ Sapienza University of Rome *05/2025 – 03/2026*
Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models
- **Simone Antonelli** – ML Research Intern @ Amboss Technologies *10/2024 – 07/2026*
Predicting Link Closure in the Lightning Network with Temporal Graph Networks
- **Marco Pegoraro** – ML Research Intern @ VantAI *10/2024 – 04/2025*
Neo 1: a unified model for all-atom structure prediction and generation of all biomolecules
- **Harrison Rush** – ML Engineer @ Amboss Technologies *08/2024 – Present*
MPFlow: Learning Budgeted Max-Flow Optimization on the Lightning Network with Deep Graph Reinforcement Learning
- **Thomas Castiglione** – ML Researcher @ VantAI *03/2024 – 04/2025*
Neo 1: a unified model for all-atom structure prediction and generation of all biomolecules
- **Andy Huang** – PhD Student @ Mila *03/2024 – 04/2025*
TGB: Temporal Graph Benchmark for Machine Learning on Temporal Graphs
UTG: Towards a Unified View of Snapshot and Event Based Models for Temporal Graphs
- **Vincent Davis** – ML Engineer @ Amboss Technologies *11/2023 – 05/2024*
Channel Balance Interpolation in the Lightning Network via Machine Learning

Research Grants & Fellowships

- **ISCRA-C Research Computing Grant** – CINECA *02/07/2026*
DOLPH-ID: Dolphin Emitter Identification from Whistle Acoustics *≈€20,000 equivalent*
Allocation of 8,500 GPU-hours on Leonardo's NVIDIA A100 64 GB accelerators. *9 months*
- **Research Fellowship** – Sapienza University of Rome *01/04/2026*
For postdoctoral research on machine learning for non-human communication. **€36,000**
12 months
- **Google Cloud Research Credits** – Google *2022, 2023*
GCP credits for PhD research. Received in two consecutive years. **\$1,000/year**

Awards

- **Olav Beckmann Project Prize** – Imperial College London *2017*
For outstanding second-year project work.
- **Engineering Dean's List** – Imperial College London *2016, 2017*
Top 10% of cohort. Received in two consecutive years.
- **G-Research Prize (Gloucester Research Ltd Prize at the time of award)** – *2016*
Imperial College London
Awarded annually to up to 10 non-final-year Computer Science students for academic excellence.

- **President's Undergraduate Scholarship** – Imperial College London 22/05/2015
Imperial's most prestigious undergraduate distinction, awarded to fewer than 5% of the incoming cohort for academic excellence and potential. Funding: £3,000.

Invited Talks

- *Graph Machine Learning for Dynamic Financial Networks: From Foundations to the Lightning Network* 10/07/2026
[Swissquote Workshop on Graph Learning in Financial Networks, EPFL](#) [[Slides \(HTML\)](#)] [[Slides \(PDF\)](#)]
- *Machine Learning on Temporal Graphs: From Foundations to Emerging Frontiers* 19/05/2026
Linguae-ML Seminar, Ecole Normale Supérieure (Paris) [[Slides \(HTML\)](#)] [[Slides \(PDF\)](#)]
- *Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models* 30/01/2026
Earth Species Project Community Deep Dive [[Slides \(PDF\)](#)] [[Video](#)]
- *A Primer on Graph Neural Networks* 22/11/2023
COLT Seminar, University of Pompeu Fabra, Barcelona [[Slides \(PDF\)](#)]
- *Edge Directionality Improves Learning on Heterophilic Graphs* 14/09/2023
Temporal Graph Reading Group [[Slides \(PDF\)](#)] [[Video](#)]
- *Edge Directionality Improves Learning on Heterophilic Graphs* 07/09/2023
[Explainability and Applicability of GNNs Workshop, Kassel](#)
- *Improving Machine Learning at Twitter using Graphs* 28/09/2022
[Privacy Week](#) [[Video](#)]
- *Learning on Graphs with Missing Node Features* 09/09/2022
[TigerGraph Reading Group](#)
- *Learning on Graphs with Missing Node Features* 01/03/2022
[Learning on Graphs and Geometry Reading Group](#) [[Slides \(PDF\)](#)] [[Video](#)]
- *Learning on Graphs with Missing Node Features* 24/02/2022
[Machine Learning Research Group, University of Oxford](#)
- *Learning on Graphs with Missing Node Features* 23/11/2021
[University of Cambridge AI Research Group Talks](#)
- *Graph Neural Networks with Almost No Features* 17/11/2021
[Toronto Machine Learning Summit](#)
- *Machine Learning on Dynamic Graphs: Temporal Graph Networks* 16/06/2021
[MLOps World 2021](#)
- *Machine Learning on Dynamic Graphs: Temporal Graph Networks* 26/04/2021
Intel ML Seminar [[Slides \(PDF\)](#)]
- *Machine Learning on Dynamic Graphs: Temporal Graph Networks* 09/04/2021
[GNNsSys'21 Workshop](#)
- *Dynamic Graphs and Temporal Graph Networks* 22/01/2021
[University of Liverpool Networks and Distributed Computing Series](#) [[Slides \(PDF\)](#)]
- *TGN: Temporal Graph Networks* 21/07/2020
[AISC](#) [[Slides \(PDF\)](#)] [[Video](#)]

Academic Service

- **Post-AGI Science and Society Workshop** – ICLR 2026 04/2026
PC Member
- **Temporal Graph Learning Workshop** – NeurIPS 2023 12/2023
Organizer
- **Temporal Graph Learning Workshop** – NeurIPS 2022 12/2022
Panel

- **NeurReps Workshop – Symmetry and Geometry in Neural Representations –** 12/2022
NeurIPS 2022
PC Member
- **GCLR 22 – 2nd Workshop on Graphs and more complex structures for learning** 02/2022
and reasoning – AAAI 2022
PC Member
- **GReS – Workshop on Graph Neural Networks for Recommendation and Search** 10/2021
– ACM RecSys 2021
PC Member
- **GNNSys’21 – Workshop on Graph Neural Networks and Systems** – MLSys 2021 04/2021
PC Member
- **AAAI-21 GCLR** – AAAI 2021 02/2021
PC Member

Full Publication List

Under Review

- F. Mustun, C. Semenzin, R. Dessì, P. R. Guerrero, P. Orhan, A. Emanuelli, **E. Rossi**, Y. Lakretz, G. G. de Polavieja and G. Sumbre. 2026. OpenWhistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. *Under Review*. (0).
- H. Rush, V. Davis, S. Antonelli, V. Singh, J. Shrader and **E. Rossi**. 2026. MPFlow: Learning Budgeted Max-Flow Optimization on the Lightning Network with Deep Graph Reinforcement Learning. *arXiv*. (0).
- **E. Rossi** and E. Rodolà. 2026. Communicating Sound Through Natural Language. *arXiv*. (0).
- S. Antonelli, V. Davis, H. Rush, A. Potdevin, J. Shrader, V. Singh and **E. Rossi**. 2026. Predicting Channel Closures in the Lightning Network with Machine Learning. *arXiv*. (0).
- D. Kovtun, M. Akdel, A. Goncarenco, G. Zhou, G. Holt, D. Baugher, D. Lin, Y. Adeshina, T. Castiglione, X. Wang, C. Marquet, M. McPartlon, T. Geffner, **E. Rossi**, G. Corso, H. Stärk, Z. Carpenter, E. Kucukbenli, M. Bronstein and L. Naef. 2024. PINDER: The Protein Interaction Dataset and Evaluation Resource. *bioRxiv*. (37).

Peer-reviewed journal articles

- D. Malitesta, **E. Rossi**, C. Pomo, T. Di Noia and F. D. Malliaros. 2026. Training-Free Graph-Based Imputation of Missing Modalities in Multimodal Recommendation. *IEEE Transactions on Knowledge and Data Engineering*. (2).
- Y. Aluma, Z. Baron, R. Barrett, C. Baumgartner, G. Beguš, S. Bhattacharya, M. M. Bronstein, S. Dahan, O. Davis, S. de Haas, J. Defoe, J. DelPreto, R. Dessì, R. Diamant, J. Gatesy, K. George, S. Gero, D. Gibbons, D. Gibbons, S. Gil, S. Goldwasser, D. F. Gruber, O. Harve, A. Hernandez, M. Ishay, N. Jadhav, L. KC, A. Kenny, A. Leitao, M. Lucas, A. Maalouf, P. Malkin, Y. Mevorach, S. Pagani, O. Paradise, G. Petri, S. Poetto, **E. Rossi**, D. Rus, M. Salino-Hugg, A. Santoro, P. Sharma, D. Tchernov, A. Torralba, P. Tønnesen, D. M. Vogt and R. J. Wood. 2026. Description of a collaborative sperm whale birth and shifts in coda vocal styles during key events. *Scientific Reports*. (1).
- V. Singh, M. Khanzadeh, V. Davis, H. Rush, **E. Rossi**, J. Shrader and P. Lio. 2025. Bayesian Binary Search. *Algorithms*. (6).

Peer-reviewed conference proceedings

- V. Davis, V. Singh and **E. Rossi**. 2025. Channel Balance Interpolation in the Lightning Network via Machine Learning. *IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*. (4).
- J. Gastinger, S. Huang, M. Galkin, E. Loghmani, A. Parviz, F. Poursafaei, J. Danovitch, **E. Rossi**, I. Koutis, H. Stuckenschmidt, R. Rabbany and G. Rabusseau. 2024. TGB 2.0: A Benchmark for Learning on Temporal Knowledge Graphs and Heterogeneous Graphs. *Advances in Neural Information Processing Systems*. (47).
- D. Malitesta, **E. Rossi**, C. Pomo, T. Di Noia and F. D. Malliaros. 2024. Do We Really Need to

Drop Items with Missing Modalities in Multimodal Recommendation?. *Proceedings of the 33rd ACM International Conference on Information.* (50).

- S. Huang, F. Poursafaei, R. Rabbany, G. Rabusseau and **E. Rossi**. 2024. UTG: Towards a Unified View of Snapshot and Event Based Models for Temporal Graphs. *Learning on Graphs Conference (LoG)*. (20).

- S. Huang, F. Poursafaei, J. Danovitch, M. Fey, W. Hu, **E. Rossi**, J. Leskovec, M. M. Bronstein, G. Rabusseau and R. Rabbany. 2023. Temporal Graph Benchmark for Machine Learning on Temporal Graphs. *Advances in Neural Information Processing Systems*. (273).

- **E. Rossi**, B. Charpentier, F. Di Giovanni, F. Frasca, S. Günnemann and M. M. Bronstein. 2023. Edge Directionality Improves Learning on Heterophilic Graphs. *Learning on Graphs Conference (LoG)*. (202).

- B. Chamberlain, S. Shirobokov, **E. Rossi**, F. Frasca, T. Markovich, N. Hammerla, M. M. Bronstein and M. Hansmire. 2022. Graph Neural Networks for Link Prediction with Subgraph Sketching. *International Conference on Learning Representations (ICLR)*. **Oral (top 5%)**. (230).

- M. Mutti, R. De Santi, **E. Rossi**, J. F. Calderon, M. M. Bronstein and M. Restelli. 2022. Provably Efficient Causal Model-Based Reinforcement Learning for Systematic Generalization. *Proceedings of the AAAI Conference on Artificial Intelligence*. (36).

- **E. Rossi**, F. Monti, Y. Leng, M. M. Bronstein and X. Dong. 2022. Learning to Infer Structures of Network Games. *Proceedings of the 39th International Conference on Machine Learning, ICML*. (13).

- **E. Rossi**, H. Kenlay, M. I. Gorinova, B. Chamberlain, X. Dong and M. M. Bronstein. 2022. On the Unreasonable Effectiveness of Feature propagation in Learning on Graphs with Missing Node Features. *Learning on Graphs Conference (LoG)*. (196).

- B. Chamberlain, J. Rowbottom, M. I. Gorinova, S. D. Webb, **E. Rossi** and M. M. Bronstein. 2021. GRAND: Graph Neural Diffusion. *Proceedings of the 38th International Conference on Machine Learning, ICML*. (544).

- B. Chamberlain, **E. Rossi**, D. Shiebler, S. Sedhain and M. Bronstein. 2020. Tuning Word2vec for Large Scale Recommendation Systems. *RecSys - 14th ACM Conference on Recommender Systems*. (38).

Peer-reviewed workshop proceedings

- D. Marincione, D. Crisostomi, R. Dessì, E. Rodolà and **E. Rossi**. 2025. Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models. *NeurIPS Workshop on AI for Animal Communication*. **Spotlight (top 15%)**. (3).

- J. Durairaj, Y. Adeshina, Z. Cao, X. Zhang, V. Oleinikovas, T. Duignan, Z. McClure, X. Robin, G. Studer, D. Kovtun, **E. Rossi**, G. Zhou, S. Veccham, C. Isert, Y. Peng, P. Sundareson, M. Akdel, G. Corso, H. Stärk, G. Tauriello, Z. Carpenter, M. Bronstein, E. Kucukbenli, T. Schwede and L. Naef. 2024. PLINDER: The Protein-Ligand Interactions Dataset and Evaluation Resource. *ICML ML for Life and Material Science Workshop*. (95).

- **E. Rossi**, B. Chamberlain, F. Frasca, D. Eynard, F. Monti and M. Bronstein. 2020. Temporal Graph Networks for Deep Learning on Dynamic Graphs. *ICML Workshop on Graph Representation Learning*. (1747).

- **E. Rossi**, F. Frasca, B. Chamberlain, D. Eynard, M. Bronstein and F. Monti. 2020. SIGN: Scalable Inception Graph Neural Networks. *ICML Workshop on Graph Representation Learning*. (600).

- **E. Rossi**, F. Monti, M. Bronstein and P. Lio. 2019. ncRNA Classification with Graph Convolutional Networks. *KDD Workshop on Deep Learning on Graphs*. (40).